DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026 : WORKSHEET-1

Subject: CHEMISTRY

Chapter: 1 –CHEMICAL REACTIONS AND EQUATIONS

Date: _____ Name : _____ Class / Sec. X_____

1. What is a chemical reaction?

   (a) A process that results in the formation of a new substance

   (b) A process where no new substance is formed

   (c) A process where two substances are mixed

   (d) A physical change in matter

2. Which of the following is a chemical reaction?

   (a) Melting wax (b) Burning a candle

   (c) Freezing water (d) Dissolving sugar in water

3. What is produced when a fuel burns in oxygen?

   (a) Salt and water (b) Oxygen and nitrogen

   (c) Hydrogen and carbon (d) Water and carbon dioxide

4. What is the type of reaction when a substance breaks down into two or more simpler substances?

   (a) Redox reaction (b) Decomposition reaction

   (c) Displacement reaction (d) Combination reaction

5. What is the role of a catalyst in a chemical reaction?

   (a) It speeds up the reaction without being consumed (b) It increases the activation energy

   (c) It is consumed in the reaction (d) It slows down the reaction

6. Which of the following is a redox reaction?

   (a) $CuSO_4 + Zn \rightarrow ZnSO4 + Cu$ (b) $NaOH + HCl \rightarrow NaCl + H2O$

   (c) $CaCO_3 \rightarrow CaO + CO2$ (d) $2Na + Cl_2 \rightarrow 2NaCl$

7. Which of the following is an example of a displacement reaction?

   (a) $AB + CD \rightarrow AD + CB$ (b) $A + B \rightarrow AB$

   (c) $A + BC \rightarrow AC + B$ (d) $AB \rightarrow A + B$

8. In the reaction between zinc and hydrochloric acid, what is the gas released?

(a) Chlorine (b) Nitrogen (c) Hydrogen (d) Oxygen

9. Burning of candle is a _____ change.

(a) Physical (b) Chemical (c) Both (a) and (b) (d) None of these

10. The reaction in which decomposition takes place after supply of heat is called _____.

(a) Thermal decomposition (b) Combination reaction

(c) Redox reaction (d) Displacement reaction

11. What is the colour of the ash formed when a magnesium ribbon is burnt in air?

(a) White (b) Black (c) Yellow (d) Pink

12. Which law should be kept in mind while we balance chemical equations?

(a) Conservation of momentum (b) Conservation of mass

(c) Conservation of energy (d) Conservation of frequency

13. Heating of ferrous sulphate is a type of:

(a) Decomposition reaction (b) Combination reaction

(c) Displacement reaction (d) All of the above

14. Rancidity can be prevented by:

(a) Adding antioxidants (b) Storing food away from light

(c) Keeping food in refrigerator (d) All of the above

15. Oxidation is a process which involves:

(i) addition of oxygen (ii) addition of hydrogen

(iii) removal of oxygen (iv) removal of hydrogen

(a) (i) and (ii) (b) (ii) and (iii) (c) (i) and (iv) (d) (ii) and (iv)

16. Which of the following statements is correct with regard to physical changes?

(a) In physical change, new substance is formed.

(b) In physical change, no new substance is formed.

(c) In physical change, chemical composition of substance is changed.

(d) None of these

17. Which of the following is not a chemical reaction?

(a) Souring of milk (b) Dissolution of sugar in water

(c) Rusting of iron (d) Digestion of food in the body

18. When white silver chloride is left exposed to sunlight, its colour becomes:

(a) Grey (b) Yellow (c) Green (d) Red

19. In which of the following, the identity of initial substance remains unchanged?

(a) Curdling of milk (b) Formation of crystals by process of crystallization

(c) Fermentation of grapes (d) Digestion of food

20. In a double displacement reaction between sodium sulphate solution and barium chloride solution:

(i) exchange of atoms takes place (ii) exchange of ions takes place

(iii) a precipitate is produced (iv) an insoluble salt is produced

The correct option is:

(a) (ii) and (iv) (b) (i) and (iii) (c) Only (ii) (d) (ii), (iii) and (iv)

21. Electrolysis of water is a decomposition reaction. The mole ratio of hydrogen and oxygen gases liberated during electrolysis of water is _____.

(a) 1 : 1 (b) 2 : 1 (c) 4 : 1 (d) 1 : 2

22. Which precipitate is formed when carbon dioxide gas is passed through lime water?

(a) Ca(OH)2 (b) Ca(OH)4 (c) CaCO3 (d) CaCO2

23. A new hammer made of iron is shiny while an old one has a reddish-brown deposit. This change indicates:

(a) Effect of moisture on metals (b) Iron hammer turns brown after some time

(c) Effects of keeping in a box for long (d) Iron changes colour when kept with other tools

24. Why do food companies fill bags of chips with nitrogen gas?

(a) It prevents rancidity of chips (b) It keeps the mosquitoes away

(c) It keeps chips dry if the pack falls in water (d) It prevents chips from spilling out

25. A student adds lead and silver to copper sulphate solution. Only lead shows a change. Why?

(a) A displacement reaction takes place (b) A combination reaction takes place

(c) Decomposition reaction occurs (d) Double displacement reaction occurs

26. When crystals of lead nitrate are heated strongly:

(a) Crystals immediately melt (b) A brown residue is left

(c) White fumes appear (d) A yellow residue is left

27. When dilute ferrous sulphate is added to acidified potassium permanganate, the purple colour fades. Why?

(a) KMnO4 is an oxidizing agent (b) FeSO4 acts as oxidizing agent

(c) Colour disappears due to dilution (d) KMnO4 decomposes

28. Which observation is correct when dilute hydrochloric acid is added to zinc?

(a) Surface of metal shines
(b) Reaction mixture turns milky
(c) Pungent gas recorded
(d) Colourless and odourless gas evolved

29. When a magnesium ribbon is burnt in air, the ash formed is:

(a) Black
(b) White
(c) Yellow
(d) Pink

30. Which observation is correct when NaOH, anhydrous CuSO4 and NaCl are added to water?

(a) Only I
(b) Only II
(c) I and IV
(d) II and III

31. What is required to identify the gas evolved when HCl reacts with zinc?

(a) Red litmus paper
(b) pH paper
(c) Lime water
(d) A burning splinter

32. On immersing an iron nail in CuSO4 solution:

(a) No reaction
(b) Colour fades
(c) Iron nail acquires a black coating
(d) Colour changes to green

33. What happens when ferrous sulphate crystals are heated?

(a) Gas with smell of burning sulphur is evolved
(b) No gas evolved
(c) Brown coloured gas evolved
(d) Colourless and odourless gas evolved

34. What is the colour of the precipitate when barium chloride solution reacts with sodium sulphate?

(a) Blue
(b) Black
(c) White
(d) Green

35. How does the colour change when gases from thermal decomposition of ferrous sulphate react with acidified potassium dichromate?

(a) Green to orange
(b) Red to colourless
(c) Orange to green
(d) Blue to green

36. What type of chemical reaction is represented by the reaction of calcium oxide with water?

(a) Combination reaction
(b) Decomposition reaction
(c) Displacement reaction
(d) Double displacement reaction

37. Which of the following is an example of a decomposition reaction?

(a) $2H_2O \rightarrow 2H_2 + O_2$
(b) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
(c) $C + O_2 \rightarrow CO_2$
(d) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

38. Which substance is reduced in the following reaction? $ZnO + C \rightarrow Zn + CO$

(a) ZnO
(b) C
(c) Zn
(d) CO

39. Which is the correct balanced form of the reaction: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$

(a) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
(b) $4Fe + 3H_2O \rightarrow Fe_3O_4 + 3H_2$
(c) $3Fe + 2H_2O \rightarrow Fe_3O_4 + 2H_2$
(d) $2Fe + 3H_2O \rightarrow Fe_3O_4 + H_2$

40. Which of the following statements is true about a chemical reaction?

(a) Energy is always absorbed
(b) Only physical change occurs
(c) A new substance is formed
(d) No change in mass occurs

41. The brown colour of iron object changes to reddish brown when it:

(a) Is exposed to dry air
(b) Reacts with acids
(c) Rusts
(d) Burns

42. What happens when quicklime is added to water?

(a) Absorbs heat
(b) Emits heat
(c) No reaction
(d) Forms an acid

43. The reaction between AgNO3 and NaCl is an example of:

(a) Combination reaction
(b) Decomposition reaction
(c) Double displacement reaction
(d) Redox reaction

44. Which of these is a redox reaction?

(a) $H_2 + Cl_2 \rightarrow 2HCl$
(b) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
(c) $NaOH + HCl \rightarrow NaCl + H2O$
(d) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

45. A reaction that absorbs heat is called:

(a) Exothermic
(b) Endothermic
(c) Thermonuclear
(d) Spontaneous

46. In the reaction $PbO + C \rightarrow Pb + CO$, which is the reducing agent?

(a) PbO
(b) C
(c) Pb
(d) CO

47. Which is a decomposition reaction caused by electricity?

(a) Electrolysis of water
(b) Heating limestone
(c) Rusting of iron
(d) Photosynthesis

48. Which of these reactions is endothermic?

(a) Formation of H_2O
(b) Electrolysis of water
(c) Neutralization reaction
(d) Burning of coal

49. Which chemical reaction shows precipitation?

(a) $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
(b) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
(c) $2H_2 + O_2 \rightarrow 2H_2O$
(d) $CaO + H_2O \rightarrow Ca(OH)_2$

50. What is the color of ferrous sulfate crystals?

(a) Blue
(b) Green
(c) White
(d) Yellow

51. Which is NOT a type of chemical reaction?

(a) Combination (b) Decomposition (c) Displacement (d) Sublimation

52. A reaction in which two compounds exchange ions to form two new compounds is called:

(a) Decomposition reaction (b) Double displacement reaction

(c) Combination reaction (d) Displacement reaction

53. Which among the following is an oxidizing agent?

(a) Hydrogen (b) Carbon (c) Oxygen (d) Zinc

54. In the reaction between zinc and hydrochloric acid, the oxidized species is:

(a) Zn (b) HCl (c) H_2 (d) $ZnCl_2$

55. Which of these reactions is NOT a redox reaction?

(a) Burning of magnesium (b) Rusting of iron

(c) Decomposition of calcium carbonate (d) Reaction between H_2 and Cl_2

56. When silver chloride is exposed to sunlight, it turns:

(a) Green (b) Brown (c) Black (d) Yellow

57. What is the chemical reaction called when an element displaces another from its compound?

(a) Double displacement (b) Decomposition (c) Combination (d) Displacement

58. The reaction: $CuSO4 + Fe \rightarrow FeSO4 + Cu$ is an example of:

(a) Combination (b) Double displacement (c) Decomposition (d) Displacement

59. The product formed when zinc reacts with copper sulfate solution is:

(a) $CuSO_4$ (b) $Zn\ SO_4$ (c) $ZnCO_3$ (d) Cu_2O

60. In which reaction does a single substance break into two or more substances?

(a) Combination reaction (b) Decomposition reaction

(c) Displacement reaction (d) Redox reaction

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